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PAPER_1087: Time-Evolving Dark Energy Equation of State w_{DE} from SCm Phonon Dynamics

Abstract

We derive the time-evolving dark energy equation of state:

$$w_{\mathrm{DE}}(t,\Gamma) = -1 + \frac{2\kappa t + \frac{[\mathrm{SSq}]}{26}\,t}{\ln\left(\Phi(\Gamma)\right)}$$

where $\kappa = 5.787 \times 10^{-9}\ \mathrm{s}^{-1}$, $[\mathrm{SSq}] = 0.57$, and $\Phi(\Gamma) = \Phi_0 \cdot S_{26}$ with $\Phi_0 = 10^{20}$. At $t = 0$ the result recovers $w_{\mathrm{DE}} = -1$ (ΛCDM), while at finite t the equation of state evolves through quintessence ($w > -1$) or phantom ($w < -1$) regimes depending on the sign of $\ln(\Phi)$.

§1 Derivation from SCm Dynamics

The dark energy pressure is defined from the density (PAPER_1086):

$$p_{\mathrm{DE}} = w_{\mathrm{DE}} \cdot \rho_{\mathrm{DE}} \cdot c^2$$

Taking the logarithmic time derivative of $\rho_{\mathrm{DE}}(t,\Gamma)$ and imposing the continuity equation $\dot{\rho}+3H(1+w)\rho = 0$:

$$w_{\mathrm{DE}}(t,\Gamma) = -1 + \frac{2\kappa t + \frac{[\mathrm{SSq}]}{26}\,t}{\ln(\Phi)}$$

§2 Phantom Crossing Analysis

The phantom divide $w = -1$ is crossed when the numerator changes sign. Since $\kappa > 0$ and $[\mathrm{SSq}] > 0$, for $t > 0$:

- If $\ln(\Phi) > 0$ (i.e., $\Phi > 1$): $w > -1$ (quintessence regime)
- If $\ln(\Phi) < 0$ (i.e., $\Phi < 1$): $w < -1$ (phantom regime)

With $\Phi_0 = 10^{20}$ and $S_{26} \sim 1.86$, $\ln(\Phi) \approx 46.7$, placing the SCm theory firmly in the quintessence regime for $t > 0$.

$$\Delta w(t) = w_{\mathrm{DE}}(t) - w_{\Lambda\mathrm{CDM}} = \frac{2\kappa t + \frac{[\mathrm{SSq}]}{26}\,t}{\ln(\Phi)}$$

§3 Time Evolution Table

Epoch t (Gyr)	w_{DE}	$ \Delta w$	Regime
0	-1.0000	0	ΛCDM recovery
1	-0.9959	0.0041	Quintessence
5	-0.9795	0.0205	Quintessence
10	-0.9591	0.0409	Quintessence
13.8	-0.9435	0.0565	Quintessence (present)

§4 Observational Signatures

The deviation $\Delta w \sim 0.06$ at $t = 13.8$ Gyr is within reach of Stage IV dark energy experiments (DESI, Euclid, Roman).

The SCm prediction is:

$$w_{\mathrm{DE}}(t_0) = -1 + \frac{2 \times 5.787 \times 10^{-9} \times 4.35 \times 10^{17} + \frac{0.57}{26} \times 4.35 \times 10^{17}}{46.7} \approx -0.94$$

References

- PAPER_1086: SCm Dark Energy Density with Γ-Coupled Phonon Modulation
- PAPER_1076: SCm Dark Energy with Phonon Linewidth Γ-Modulation
- PAPER_889: EtVsLambdaCDMDarkEnergyContrastCalc

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